

[NJC/2024/Q3]

3 Name your Jupyter Notebook as:

TASK3_<your name>.ipynb

The task is to:

- read a file containing a list of numbers
- store them into a list
- sort the list using a merge sort
- choose n random elements from the list
- write the list of random elements to a file
- compute the average of these n random elements
- find the number of elements in the list that is less than average of these n random elements

For each of the sub-tasks, add a comment statement at the beginning of the code, using the hash symbol '#' to indicate the sub-task the program code belongs to, for example:

```
In [1]: #Task 3.1  
        Program Code
```

Output:

Task 3.1

Write a function task3_1(list_of_values) that:

- takes a list of values, list_of_values
- sorts them into ascending order using merge sort
- returns the sorted list.

[NJC/2024/Q3 Cont']

Use the list [56, 25, 4, 98, 0, 18, 4, 5, 7, 0] to test your function.

For example, the condition

`task3_1([56, 25, 4, 98, 0, 18, 4, 5, 7, 0]) == [0, 0, 4, 4, 5, 7, 18, 25, 56, 98]`

should return True.

[2]

Task 3.2

Write a function `task3_2(filename_in, filename_out, no_of_sample)` that :

- accepts three parameters:
 - `filename_in`, a string representing the input file name
 - `filename_out`, a string representing the output file name
 - `no_of_sample`, an integer representing the number of random elements to take from the list
- read the values from the input file `filename_in`
- store them into a list
- choose `no_of_sample` random elements from the list
- writes those values, one per line, to a file named `filename_out`
- calculates the arithmetic average of the `no_of_sample` random elements taken
- return the number of values found in file `filename_in` that is less than arithmetic average of the values found in `filename_out`

[6]

Task 3.3

The collection of values of `no_of_sample` elements taken from the file `TASK3FILE.txt` is called lower-sufficient if the number of values in the file `TASK3FILE.txt` that is less than the average of the values in the collection, is less than an integer α . Else, the collection is called lower-insufficient.

Let `no_of_sample` = 700 and α = 1000.

(a) Find a lower-insufficient collection and write it to the file `NOTLOWER.TXT`,

(b) Find a lower-sufficient collection and write it to the file `LOWER.TXT`.

[4]

Save your Jupyter Notebook for Task 3.